

Muzammal NASEER

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EDUCATION

- 2017 - 2020 | **PhD** - Engineering & Computer Science
Australian National University (ANU)
Thesis: "Analysing & Improving Neural Network's Robustness against Adversarial Perturbations."
- 2011 - 2013 | **Master** of Science in ELECTRICAL ENGINEERING
King Fahd University of Petroleum & Minerals (KFUPM)
Thesis: "The Design of Finite Impulse Response (FIR) Wavefield Extrapolation Filters."
GPA: 3.75/4.0, First Class Honours.
- 2006 - 2010 | **Bachelor** of Science in ELECTRICAL ENGINEERING
University of the Punjab (PU)
Thesis: "Spectrum Monitoring & Analysis of Interference in EGSM Networks."
GPA: 3.89/4.0, First Class Honours, Gold Medalist.

WORK EXPERIENCE

- Sept. 2020 - Present | *Research Associate* **Mohamed bin Zayed University of Artificial Intelligence - MBZUAI, UAE**
Developing novel algorithms for adversarial attacks and defenses, few-shot learning and domain generalization. I am also supervising MS/Ph.D. students and research assistants.
- Sept. 2018 - Mar. 2020 | *Research Assistant* **Inception Institute of Artificial Intelligence - IIAI, UAE**
Developed novel adversarial attack and defense algorithms for deep neural networks.
- Jan. - Jul. 2018 | *Research Assistant* **Data61, CSIRO, Canberra, Australia**
Developed new black-box attack methods for transferable adversarial perturbations for neural networks based vision systems.
- Sept. 2014 - June. 2017 | *Lecturer (Full-Time)* **Department of Electrical Engineering - subcampus of KFUPM, KSA**
I joined subcampus of KFUPM, now known as "University of Hafr Al-Batin" as a full-time lecturer to teach variety of Electrical Engineering courses.
- Jan. - May 2013 | *Teaching Assistant* **Department of Electrical Engineering, KFUPM, KSA**
Assisted in Control Theory and Signal Processing courses for the period of one semester.
- May - June 2009 | *Research Intern* **Frequency Allocation Board (FAB), Pakistan**
Worked with FAB team to develop the direction finding system for RF Interferers.

RESEARCH INTERESTS

I am intrigued by brittleness of deep neural networks in (non) i.i.d settings, thus I am focused on understanding and improving out-of-distribution generalization of these models. Currently, I am developing new algorithms for "Adversarial Attack and Defense" methods, "Attention based Modeling", and "Few-shot" learning. My research includes applications to classification, segmentation, object detection and visual tracking.

PUBLICATIONS

[GOOGLE SCHOLAR](#)

- M. Naseer, S. Khan, M. Hayat, F. Shahbaz, MH Yang. "Intriguing Properties of Vision Transformers", Advances in Neural Information Processing Systems (NeurIPS-2021-Spotlight) [[Paper](#), [code](#)]
- M. Naseer, S. Khan, M. Hayat, F. Shahbaz, F. Porikli. "On Generating Target Transferable Perturbations", International Conference on Computer Vision (ICCV-2021) [[Paper](#), [code](#)]
- K. Ranasinghe, M. Naseer, S. Khan, M. Hayat, F. Shahbaz. "Orthogonal Projection Loss", International Conference on Computer Vision (ICCV-2021) [[Paper](#), [code](#)]
- M. Naseer, S. Khan, M. Hayat, F. Shahbaz, F. Porikli. "A Self-supervised Approach for Adversarial Robustness", IEEE Conference on Computer Vision and Pattern Recognition (CVPR-2020-Oral) [[paper](#), [Presentation](#), [Poster](#), [code](#)].
- M. Naseer, S. Khan, H. Khan, F. Shahbaz, F. Porikli. "Cross-Domain Transferability of Adversarial Perturbations", Advances in Neural Information Processing Systems (NeurIPS'2019) [[paper](#), [poster](#), [code](#)]
- M. Naseer, S. Khan, F. Porikli. "Local Gradients Smoothing: Defense against localized adversarial attacks", Winter Conference on Applications of Computer Vision (WACV'2019) [[paper](#)]

PUBLICATIONS

M. Afham, S. Khan, MH Khan, **M. Naseer**, F. Shahbaz. “*Rich Semantics Improve Few-shot Learning*”, The British Machine Vision Conference (BMVC-2021) [[Paper](#), [code](#)]

M. Naseer, S. Khan, F. Porikli. “*Indoor Scene Understanding in 2.5/3D for Autonomous Agents: A Survey*”, Multidisciplinary Open Access Journal (IEEE Access’2019) [[paper](#)]

M. Naseer, W. Mousa. “*Linear complementarity problem: A novel approach to design finite-impulse response wavefield extrapolation filters*”, Society of Exploration Geophysicists. (GEOPHYSICS’2014) [[paper](#)]

M. Naseer, W. Mousa, T. Al-Naffouri, A. Al-Shuhail. “*One Dimensional Wavefield Extrapolation Filter Design Via L1 Error Approximation*”, SEG Technical Program Expanded Abstracts (GEOPHYSICS’2015) [[paper](#)]

M. Naseer, K. Ranasinghe, S. Khan, F. Shahbaz, F. Porikli. “*On Improving Adversarial Transferability of Vision Transformers*”, (arXiv-2021) [[preprint](#), [code](#)]

S. Khan, **M. Naseer**, M. Hayat, SW. Zamir, F. Shahbaz Khan, Mubarak Shah. “*Transformers in vision: A survey*”, (arXiv-2021) [[preprint](#)]

M. Naseer, S. Khan, M. Hayat, F. Shahbaz, F. Porikli. “*Stylized Adversarial Defense*”, (arXiv-2020) [[preprint](#), [code](#)]

HONORS AND AWARDS

DEC. 2019 **Student Travel Award** by Conference on Neural Information Processing Systems (NeurIPS-2019).

JUNE 2017 PHD: Awarded **Postgraduate Research Scholarship** by Australian National University for the period of three years.

JUNE 2017 PHD: Awarded an ANU HDR **Fee Remission Merit Scholarship** by Australian National University for the period of four years.

JAN. 2011 M.Sc: Awarded Fully Funded **Master of Science Scholarship** by Ministry of Higher Education, KSA for the period of two and half years.

JAN. 2011 B.Sc: Awarded **Gold Medal** for outstanding performance and first class honors in B.Sc. Electrical Engineering.

FEB. 2007-09 B.Sc: Received **University Merit Scholarship** consistently for three years for best student performance by the University of the Punjab.

COMPUTATIONAL SKILLS

PYTHON EXPERT KNOWLEDGE- I am extensively using python to build novel machine learning algorithms for the last few years.

PYTORCH Pytorch is usually my default choice due to its dynamic nature and object-oriented graph design approach.

KERAS I have used Keras with Tensorflow as back end to build different projects including a defense algorithm against unconstrained adversarial attacks. [[paper](#)]

MATLAB Most of my B.Sc and M.Sc academic research published or otherwise is implemented in Matlab.

C/C++ I scored A+ grades in these languages in my B.Sc courses. [[Transcript](#)]

REFERENCES

Dr. Salman Khan

Assistant Professor at the Mohamed bin Zayed University of Artificial Intelligence (MBZUAI),

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Dr. Munawar Hayat

Senior Lecturer at Monash University

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Dr. Fahad Shahbaz Khan

Associate Professor at Linkoping University, Sweden

Associate Professor at the Mohamed bin Zayed University of Artificial Intelligence (MBZUAI),

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Prof. Fatih Porikli

Chief Scientist at Qualcomm, United States,

Professor at Research School of Engineering, Australian National University (ANU).

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